

Project Design Phase-II - Data Flow Diagrams & User Stories

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	4 Marks

1. Data Flow Diagrams (DFD)

Level 0 DFD: Context Diagram



Level 1 DFD: Subprocess Flow

- **Process 1.0 (Auth Service):** Accepts customer details, queries MongoDB, verifies credentials, and releases JWT.
- **Process 2.0 (Product Catalog Service):** Fetches, filters (price range, categories), and returns matched products list.
- **Process 3.0 (Order & Billing Service):** Parses cart, counts local pricing, applies GST (18%), checks stock, and stores new orders.

2. User Stories Table

User Type	Epic	Story ID	User Story / Task	Acceptance Criteria	Priority
Customer	Auth	US-101	As a customer, I want to sign up with my email so I can place orders.	- Email validation required. - Password must be stored as hashed. - Return JWT upon success.	High

User Type	Epic	Story ID	User Story / Task	Acceptance Criteria	Priority
Customer	Catalog	US-201	As a customer, I want to filter products by category and max price.	- Filter sidebar loads options dynamically. - Sliders change maximum price range in real-time.	High
Customer	Cart	US-301	As a customer, I want to modify my cart item quantities without losing them on page refresh.	- Save states to LocalStorage. - Block additions exceeding stock limits.	High
Admin	Catalog	US-401	As an admin, I want to add new products to the catalog.	- Secure admin dashboard route. - Image upload URLs verified. - Real-time update in db.	High